

Validating and Extending a Chapter-Aligned Moodle Engagement Metric Across STAT0002 and STAT0004

Evidence from online, hybrid and in-person Moodle delivery cohorts

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Abstract

Virtual learning environments record how students interact with course materials, offering a scalable basis for monitoring online behavioural engagement. Johnston et al. (2025) developed a chapter-aligned real-time metric and validated it on structured lecture modules; this report applies that fixed measure to five STAT0002 and STAT0004 cohorts at University College London—spanning online, hybrid and in-person periods not used in the original study—and asks whether the association travels, what kind of engagement the components reveal, and how practitioners should use the signal safely. Cumulative engagement remains moderately associated with final grade when cohorts are pooled ($r = 0.32$), with the association concentrated among lower-performing students but modest early-warning precision (about 15–26%). Trajectory classes separate outcomes by roughly 9.7 grade points; partial weekly profiles recover the low-steady class with about 81% recall by week 4, though precision for actual failure remains similar to the quintile flag. Component decomposition shows breadth and promptness of resource use outweigh session frequency. Delivery-period comparisons are descriptive because delivery mode is confounded with cohort and academic year. The metric suits low-cost supportive monitoring with local validation, not high-stakes prediction.

1 Introduction

Student engagement—the time and effort students devote to learning activities—is widely linked to academic performance, satisfaction and retention. In online and blended higher education, much of the *behavioural* dimension of engagement can be observed through virtual learning environment (VLE) log data: which resources students access, how often they return, and how promptly they engage with newly released material. Moodle and similar platforms therefore provide an unobtrusive, scalable record of online study behaviour that is increasingly used in learning analytics and early student support (Henrie et al., 2015).

Institutions nonetheless face a practical tension. Threshold-based early-warning tools depend on cut-offs that are difficult to justify in advance, while predictive models often require historical outcome data and may fail to transfer across cohorts or courses. An alternative is to construct interpretable composite indicators grounded in engagement theory and computable in real time from log data alone. Johnston et al. (2025) developed such a measure by reconceptualising the course-wide engagement score of Chong and Wong (2019) into a weekly, chapter-aligned formulation based on Frequency, Immediacy and Diversity (F, I, D). Their study showed that the metric aligns with the retrospective course-wide score from early in the term, relates to final grades in structured lecture-based modules, and can help identify low-performing students through engagement quintiles—though with the modest precision typical of early-warning systems.

Johnston et al. (2025) establish that a chapter-aligned engagement metric can be computed in real time and can relate to final grades in structured modules. The present report takes that metric as fixed and asks a different question: how informative, robust and interpretable it is when applied to additional STAT0002 and STAT0004 cohorts spanning online, hybrid and in-person delivery periods? The contribution is therefore not metric development, but secondary validation and interpretive extension: pooling evidence across cohorts, testing whether the association is concentrated among lower-performing students, decomposing the signal into Frequency, Immediacy and Diversity, and examining whether temporal regularity and engagement trajectories add practical interpretability.

The report is organised around three lines of evidence. First, does the metric **travel** to new cohorts—a moderate, positive engagement–grade association that holds when cohorts are pooled? Second, what **kind** of engagement matters beyond raw volume—breadth, promptness, regularity and sustained trajectory shape? Third, how should practitioners use the metric **safely**—recognising lower-tail relevance but modest precision, and treating delivery-period comparisons as descriptive boundary checks rather than causal tests. The sections below develop this argument in three stages: external validation, diagnostic interpretation, and practical limits on student support.

2 Background and metric context

Online behavioural engagement refers to observable, sustained interaction with learning activities in a digital environment. Engagement theory distinguishes behavioural, emotional and cognitive forms of involvement; Moodle logs capture primarily the behavioural dimension—participation, effort and attention to academic tasks—rather than motivation or understanding (Fredricks et al., 2004; Henrie et al., 2015). It is only a *proxy* for learning: logs record what happens inside the VLE, not offline study, in-person attendance or the quality of students’ reasoning. Even so, when behavioural engagement is measured coherently and validated against academic outcomes, it can support timely monitoring and targeted support.

The course-wide metric of Chong and Wong (2019) combines five indicators—Immediacy, Frequency, Diversity, Recency and Interval—into a single retrospective score aggregated across an entire course. That design is useful for end-of-term analysis but poorly suited to weekly monitoring, and it treats the course as homogeneous rather than aligned with its chapter structure. Johnston et al. (2025) addressed this by computing Frequency, Immediacy and Diversity separately for each chapter at each week, min-max scaling each indicator relative to peers at that point in the term, and summing them into a chapter-level score $IDF = F + I + D$. Chapter scores are combined across released chapters to yield a cumulative engagement measure that can be updated as the term progresses without outcome data or model training. Recency and Interval were excluded because they become unstable when measured mid-course.

Chapter alignment matters because engagement is most interpretable when measured in step with how material is released. For early-warning use, practitioners also need to understand the recall–precision trade-off: flagging the lowest-engagement students may capture many who ultimately struggle, but also many who would have succeeded without intervention (Conijn et al., 2017). The present report takes the metric definition as given and focuses on how it behaves on new UCL cohorts and what additional diagnostic insight can be extracted from its components and related temporal patterns.

3 Data

The analysis uses five module-year deliveries from two undergraduate statistics modules at University College London. STAT0002 is a first-year foundations module represented here by an online cohort (2020–21) and an in-person cohort (2022–23). STAT0004 is a second-year module and the only module in this dataset spanning all three delivery conditions: online (2020–21), hybrid (2021–22) and in-person (2022–23). Each delivery covers eleven teaching weeks with up to ten labelled chapters. Final grades are recorded on a 0–100 scale; grades below 50 were treated as low performance, and grades below 40 as failure. Students with a recorded grade of zero, typically reflecting exam absence, were excluded.

Table 1 summarises the cohorts. The main engagement–grade sample comprises $N = 1,291$ students with both engagement records and valid final grades after exclusions. The latent trajectory analysis below uses $N = 1,290$ students with complete weekly engagement profiles suitable for class assignment; the difference of one student reflects missing weekly coverage in the trajectory model, not an error in the cohort inventory.

Table 1: Cohort inventory for the main engagement–grade sample ($N = 1,291$).

Module	Year	Condition	N	Mean (SD)	% < 50	% < 40
STAT0002	2020–21	Online	342	66.8 (13.8)	9.9	1.8
STAT0002	2022–23	In-person	268	67.4 (13.0)	10.4	2.6
STAT0004	2020–21	Online	312	62.7 (12.2)	12.5	2.9
STAT0004	2021–22	Hybrid	172	71.9 (8.8)	1.7	0.0
STAT0004	2022–23	In-person	197	70.6 (11.3)	3.6	1.0
Total			1,291			

Source: author’s analysis of UCL Moodle log data.

Two design features constrain interpretation. First, the share of low-performing students varies widely across cohorts (from 1.7% to 12.5%), which limits how sharply any early-warning flag can perform in the highest-achieving deliveries. Second, delivery condition is **perfectly confounded** with academic year and cohort: online corresponds to 2020–21, hybrid to 2021–22, and in-person to 2022–23. Comparisons across conditions therefore describe different module-year contexts rather than randomly assigned delivery modes, and no causal claim about online, hybrid or in-person delivery can be supported.

The modules also differed in assessment structure. STAT0002 in-person records separate exam, participation and peer-marking components; STAT0004 deliveries include coursework and quiz marks, with a group component in the in-person year. This matters because Moodle engagement may align more closely with individual or exam-style assessment than with group, peer-marked or participation components. Where component marks were available, this report therefore analyses whether the engagement metric correlates more strongly with individually assessed work than with collaborative or peer-assessed components.

The two modules also differ in the strength of the engagement–grade link across deliveries. STAT0002 cohorts show end-of-term correlations of $r = 0.28$ – 0.41 , whereas STAT0004 ranges from $r = 0.19$ to $r = 0.30$. As interpretive context, the two modules may differ in ways that affect how well Moodle activity proxies study behaviour; several factors may contribute, though they cannot be disentangled in this observational design. STAT0002 is a first-year foundations module with substantial individually assessed exam weight, where sustained access to lecture materials and problem sheets may track preparation more directly. STAT0004 is a second-year module with greater student autonomy, more coursework and group-assessment weighting, and possibly more study away from Moodle—each of which would weaken the observable link between VLE logs and final outcomes. How tightly Moodle resources are organised by chapter may also

matter: where chapter labelling aligns less clearly with assessed work, a chapter-aligned metric would legitimately carry less outcome signal, as Johnston et al. (2025) report for less structured courses.

4 Engagement feature construction

Weekly chapter-level contribution values were available for each student. These values are stored as *per-week increments*, not as cumulative totals. For each student and week, contributions were summed across chapters to obtain weekly Frequency, Immediacy and Diversity totals. The weekly composite was then defined additively as

$$\text{IDF}_{\text{week}} = \text{freq}_{\text{week}} + \text{imm}_{\text{week}} + \text{div}_{\text{week}},$$

and cumulative measures were obtained by summing across weeks: `cum_freq`, `cum_imm`, `cum_div` and `cum_IDF`. This additive definition matches the chapter-level formulation of Johnston et al. (2025); it is not a multiplicative composite. A reconstruction check confirmed that the supplied contributions are consistent with the intended Frequency and Diversity definitions.

For extension analyses, additional timing and regularity features were derived from event logs (for example, variation in weekly activity and gaps between sessions). Latent-class mixed models were also fitted to normalised weekly cumulative IDF curves to identify distinct engagement trajectories over the term.

5 Methods

5.1 Association between engagement and final grade

Within each cohort, Spearman rank correlations were computed between cumulative engagement and final grade at each week and at the end of term. To synthesise evidence across cohorts, cohort-specific correlations were pooled using a random-effects meta-analysis, reporting the pooled correlation, confidence interval and I^2 heterogeneity statistic. Complementing rank-based evidence, a mixed-effects model was fitted with final grade as the outcome and standardised cumulative IDF as the predictor, allowing the slope to vary by cohort. This expresses the association on the grade scale in points per one standard deviation of engagement.

5.2 Timing and early-course usefulness

Weekly Spearman correlations between cumulative engagement and final grade were tracked across the term to assess how early the association emerges. For each cohort, a bootstrap procedure identified the earliest week at which the correlation was statistically indistinguishable from its end-of-term value, providing a practical sense of when the signal stabilises.

5.3 Lower-tail and early-warning analysis

Students were grouped into engagement quintiles at selected weeks, and the distribution of final grades within each quintile was examined. Quantile regression estimates the relationship between engagement and final grade at different points in the grade distribution—for example, among the lowest 10% of performers rather than only at the median—so that one can ask whether the metric matters more for students already at academic risk. Particular attention was given to the lower tail ($\tau = 0.1$) relative to the median ($\tau = 0.5$). For early-warning relevance, the bottom engagement quintile was treated as a risk flag and evaluated using the area under the ROC curve (AUC), recall and precision for identifying students scoring below 50% at the end of term.

5.4 Diagnostic decomposition of engagement

Frequency, Immediacy and Diversity are strongly correlated, so ordinary regression coefficients do not reveal which dimension carries unique information. Variance inflation factors were examined to confirm collinearity, and the LMG relative-importance decomposition was used to partition explained variance among the three indicators in a way that is stable under collinearity: it averages each predictor’s contribution to R^2 over all possible orderings in which the indicators enter the model.

5.5 Assessment-component validity

Where separate assessment marks were available, engagement was correlated with each component and Steiger’s test for dependent correlations was used to compare whether engagement aligns more strongly with individually assessed work (exams, quizzes) than with group- or peer-marked components.

5.6 Temporal regularity and trajectory analysis

Timing and regularity features were added to models already containing F, I and D to test whether *how* students engage over time adds information beyond *how much* and *how broadly* they engage. Separately, latent-class mixed models were applied to weekly cumulative IDF profiles to identify groups of students following distinct engagement trajectories and to compare their final grades. To assess operational usefulness, end-of-term class labels were compared with early assignments made by matching each student’s partial weekly profile (weeks 1– w) to the corresponding segment of the end-of-term class centroids.

5.7 Robustness checks

The main conclusions were examined under degree-programme adjustment, false-discovery-rate correction using the Benjamini–Hochberg procedure across weekly association tests, alternative bounded outcome specifications, and variation in the at-risk engagement threshold. These checks were kept brief and interpreted in the context of the primary findings rather than treated as a separate technical audit.

6 Results

The results follow the three contributions outlined in the Introduction: external validation on new cohorts, diagnostic interpretation of what kind of engagement matters, and practical limits on how the metric should be used for student support.

6.1 External validation: the metric associates with grades in new cohorts

Across all five cohorts, cumulative engagement is positively and moderately associated with final grade. The random-effects pooled rank correlation for cumulative IDF at week 11 is $r = 0.32$ (95% CI [0.25, 0.38]), with low-to-moderate heterogeneity across cohorts ($I^2 = 39\%$). Frequency, Immediacy and Diversity show similar pooled associations when analysed separately (Table 2). On the grade scale, the mixed-effects estimate corresponds to approximately 3.8 grade points per one standard deviation increase in engagement. Figure 1 shows that every cohort lies above zero, but the association is far from perfect prediction—engagement is a useful but imperfect proxy for academic performance.

Table 2: Pooled rank correlations and mixed-effects slopes at week 11.

Indicator	Pooled r [95% CI]	I^2	Slope (SE)	Moderator p
Frequency	0.28 [0.22, 0.34]	27%	3.36 (0.53)	0.92
Immediacy	0.30 [0.24, 0.37]	37%	3.63 (0.58)	0.88
Diversity	0.31 [0.24, 0.37]	33%	3.62 (0.53)	0.87
IDF	0.32 [0.25, 0.38]	39%	3.79 (0.58)	0.89

Source: author’s analysis of UCL Moodle log data.

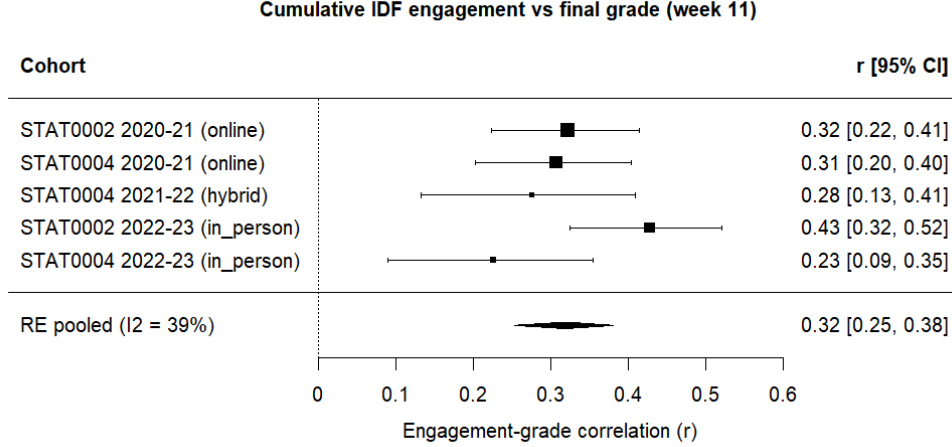


Fig. 1: Cohort-specific and pooled week-11 correlations between cumulative IDF engagement and final grade. All cohorts show positive associations; the random-effects estimate summarises the overall moderate engagement–grade relationship.

STAT0002 deliveries show stronger cohort-level associations than STAT0004 ($r = 0.28$ – 0.41 versus $r = 0.19$ – 0.30), consistent with the module-level patterns noted in Section 3. The pooled estimate nonetheless remains positive and moderate because every cohort contributes in the same direction.

When students are ranked by cumulative engagement and grouped into quintiles, final grades tend to rise across successive quintiles, particularly in STAT0002, where quintile separation is clearest (Figure 2). At weeks 3 and 6, the lowest-engagement quintile is the group whose grade distribution most often approaches or crosses the 50% low-performance threshold, while the highest-engagement quintile consistently shows the strongest outcomes. Separation between adjacent middle quintiles is weaker, which is consistent with a monotone but noisy relationship rather than sharp boundaries between moderate engagement levels. Together, the pooled correlation, positive cohort-level associations and quintile patterns support the conclusion that the metric *travels* to these new module-year contexts with a moderate, robust association.

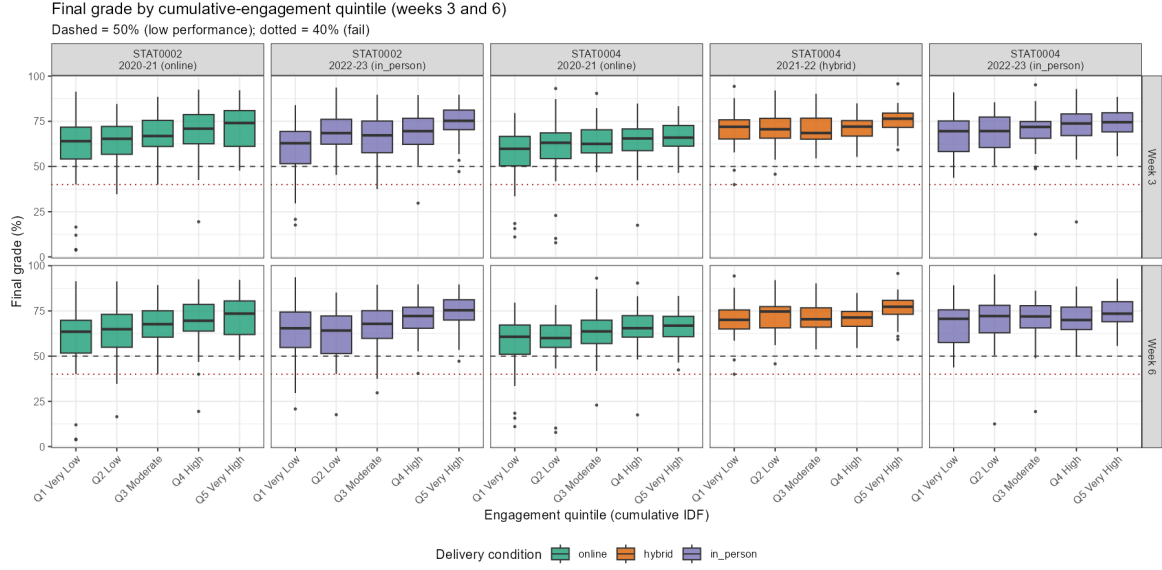


Fig. 2: Final-grade distributions by engagement quintile at weeks 3 and 6.

6.2 When the engagement–grade association stabilises

Bootstrap stabilisation analysis asks whether additional weeks materially change the engagement–grade correlation, not whether the correlation is already strong (Table 3, Figure 3). In four cohorts, the weekly correlation is statistically close to its end-of-term value from week 1 or 2, but this sometimes reflects a relatively flat and modest signal rather than impressive early predictive power. For example, STAT0004 in-person stabilises early yet has a weak end-of-term association ($r = 0.19$); STAT0004 hybrid similarly stabilises at week 1 with $r = 0.23$. By contrast, STAT0002 in-person shows a strengthening association that does not stabilise until around week 9, reaching the strongest cohort-level correlation ($r \approx 0.41$), suggesting that later engagement adds meaningful information in that delivery. Stabilisation timing and association strength should therefore be read separately: early stabilisation does not guarantee a strong early-warning signal.

Table 3: Stabilisation week and end-of-term IDF–grade correlation.

Cohort	Stabilisation week	End-of-term r
STAT0002 online	2	0.28
STAT0002 in-person	9	0.41
STAT0004 online	1	0.30
STAT0004 hybrid	1	0.23
STAT0004 in-person	1	0.19

Source: author’s analysis of UCL Moodle log data.

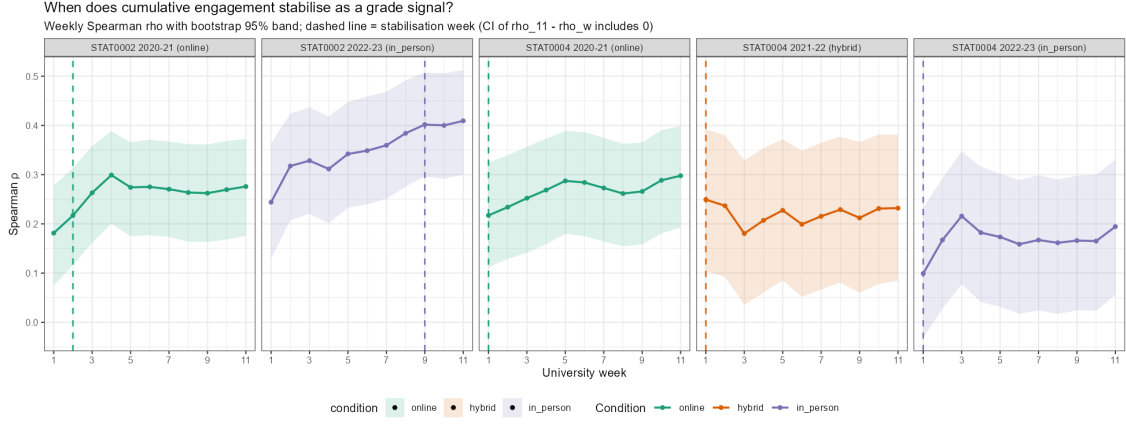


Fig. 3: Weekly Spearman correlation between cumulative engagement and final grade by cohort, with bootstrap stabilisation week marked. Early stabilisation can reflect a flat, modest signal as well as a strong one; compare stabilisation week with end-of-term r in Table 3.

6.3 Diagnostic value: trajectories, components and study behaviour

Latent-class analysis of weekly cumulative IDF profiles is the empirically strongest diagnostic result in the report. A three-class solution identifies interpretable trajectory types—low-steady, high-steady and mid-variable (Figure 4)—and final grades differ markedly across classes (Kruskal–Wallis $p = 2.6 \times 10^{-17}$). The high-steady class averages 72.9, compared with 63.2 for the low-steady class: a gap of about 9.7 grade points. The low-steady class also contains the highest share of low performers (15.7%). Sustained engagement *patterns* over the term therefore separate outcomes more sharply than a single cumulative score at one time point, and identify a group whose profile is persistently low rather than temporarily quiet.

Table 4: Trajectory classes and grade outcomes ($N = 1,290$).

Class	N	Mean grade	% < 50
Low-steady	439	63.2	15.7
High-steady	165	72.9	2.4
Mid-variable	686	68.3	5.5

Source: author’s analysis of UCL Moodle log data.

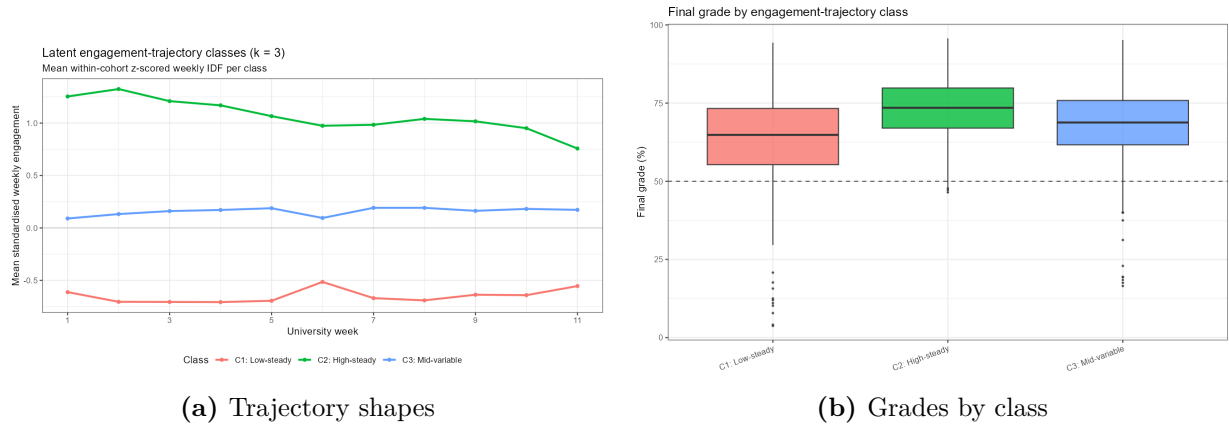


Fig. 4: Latent engagement trajectories (left) and final-grade distributions by trajectory class (right). Three trajectory types separate outcomes by roughly 9.7 grade points between the highest- and lowest-engagement profiles.

A natural operational question is how early a student can be assigned to a trajectory class with reasonable confidence. End-of-term class labels were therefore compared with early assignments made by matching each student’s partial weekly profile (weeks 1– w) to the corresponding segment of the end-of-term class centroids (Table 5, Figure 5). Overall agreement rises from 71% at week 3 to 90% by week 9. For the low-steady class specifically, recall reaches 81% by week 4 and 88% by week 6; precision that an early low-steady flag corresponds to the true end-of-term low-steady class is 77% at week 4 and 81% at week 6. These levels are materially higher than the bottom-quintile flag’s precision for identifying students who ultimately score below 50% (about 15–26%). However, among students flagged as low-steady early, only about 15% still score below 50%—similar to the quintile rule. Trajectory class is therefore a more reliable early label for *sustained low engagement* than the quintile is for *eventual failure*: operationally useful for targeted, pattern-based outreach from mid-term onwards, but not a sharp failure predictor.

Table 5: Early recovery of end-of-term trajectory class from partial weekly profiles.

Week	Agreement (%)	Low-steady recall (%)	Low-steady precision (%)	% flagged < 50
3	70.9	78.6	69.0	14.8
4	76.9	81.3	76.8	14.6
6	81.6	88.2	80.8	14.8

Source: author’s analysis of UCL Moodle log data.

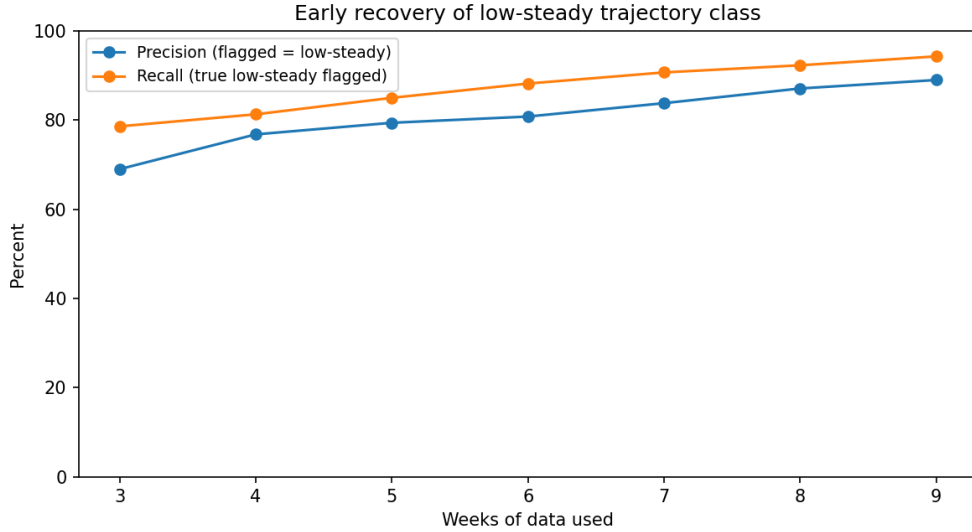


Fig. 5: Recall and precision for early assignment to the low-steady trajectory class as more weekly data accumulate. Agreement improves steadily from week 3 onwards.

Decomposing the composite into Frequency, Immediacy and Diversity clarifies *why* some engagement profiles matter more than others. Because the three indicators are highly collinear, ordinary regression coefficients are difficult to interpret; the LMG decomposition provides a clearer picture (Table 6, Figure 6). Diversity typically accounts for the largest share of explained variance (about 37% on average across cohorts), Immediacy is intermediate, and Frequency the smallest (about 27%).

This pattern is theoretically coherent. In the engagement literature, behavioural engagement is not reducible to time-on-task: it reflects participation breadth, effort and timely involvement with academic work (Fredricks et al., 2004). Frequency counts log events and is therefore sensitive to superficial or repeated access without sustained study. Diversity captures whether students explore the full range of chapter resources—a proxy for curriculum coverage rather than revisiting

a single file. Immediacy reflects alignment with the release schedule, consistent with staying current rather than falling behind or cramming at the end of term. The LMG result suggests that the metric’s outcome relevance lies in indicators of structured study behaviour, not raw click volume—the report’s strongest new insight is not merely that engagement matters, but *what kind* of engagement matters.

Table 6: LMG relative-importance shares (%) by cohort.

Cohort	Frequency	Immediacy	Diversity
STAT0002 online	28	45	27
STAT0002 in-person	28	36	36
STAT0004 online	23	32	45
STAT0004 hybrid	31	32	38
STAT0004 in-person	25	33	42

Source: author’s analysis of UCL Moodle log data.

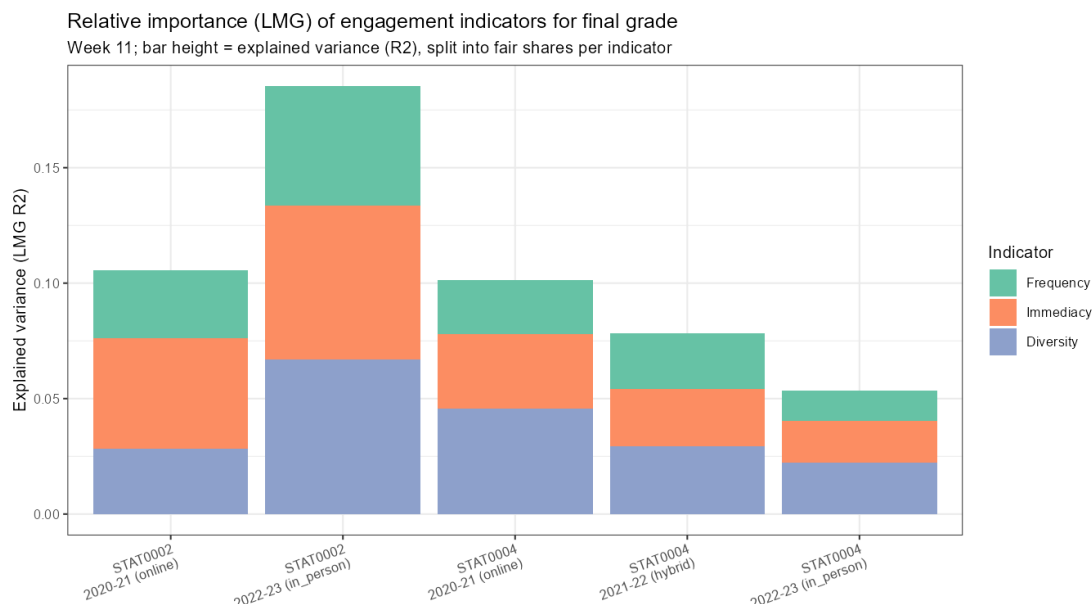


Fig. 6: LMG relative-importance shares for Frequency, Immediacy and Diversity by cohort. Diversity and Immediacy typically account for more unique explained variance than Frequency, suggesting that breadth and promptness matter more than raw activity volume.

Engagement correlates more strongly with individually assessed components than with group- or peer-marked work where both are available (Figure 7). In STAT0002 in-person, the correlation with the exam grade ($r = 0.40$) is significantly larger than with peer-marking ($r = 0.20$; Steiger $z = 2.98$, $p = 0.003$). Other component contrasts point in the same direction but are not statistically significant. This supports the interpretation that Moodle engagement reflects individual study behaviour more faithfully than outcomes shaped by group dynamics or peer assessment.

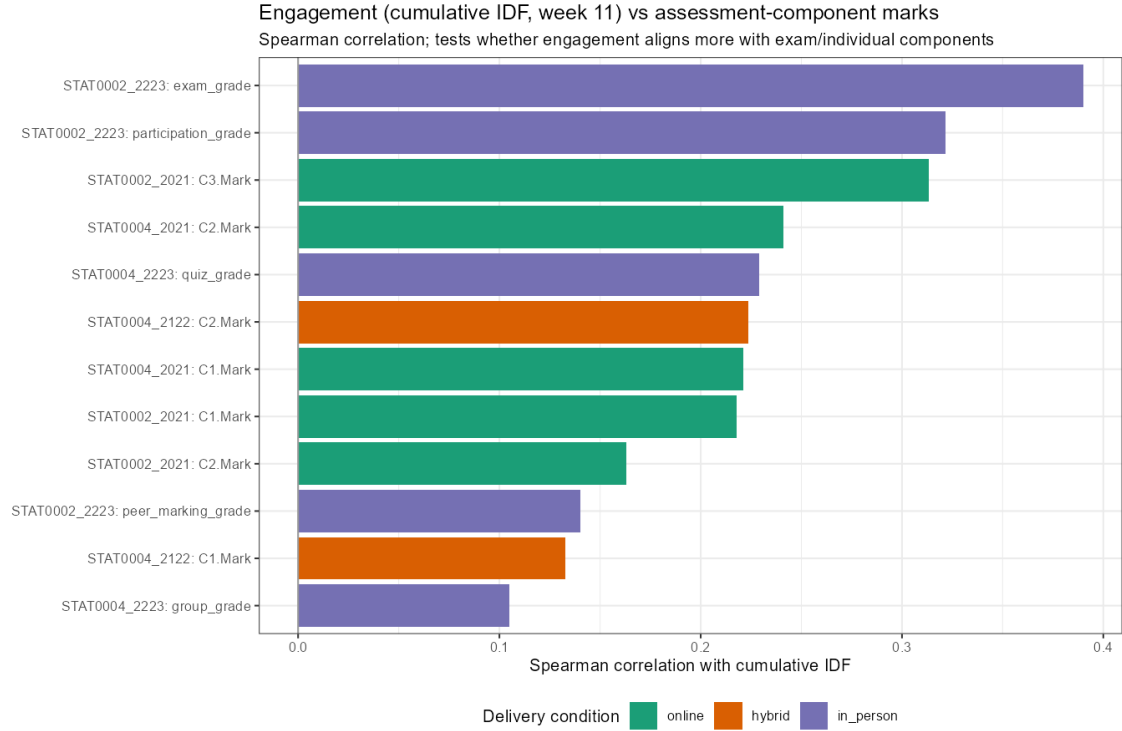


Fig. 7: Engagement–assessment component correlations by cohort.

Timing and regularity features add incremental explanatory power beyond F, I and D in the STAT0002 cohorts, where engagement showed stronger alignment with individual assessment components (Table 7, Figure 8). At week 6, the gain in R^2 is about 0.05 in both STAT0002 deliveries ($p = 0.003$ and $p = 0.006$), but not statistically significant in the STAT0004 deliveries. How regularly students work carries additional diagnostic information in STAT0002, beyond what the core composite already captures.

Table 7: Incremental R^2 from temporal features at week 6.

Cohort	R^2 (F/I/D)	R^2 (full)	ΔR^2	p
STAT0002 online	0.091	0.143	0.052	0.003
STAT0002 in-person	0.156	0.213	0.057	0.006
STAT0004 online	0.089	0.094	0.005	0.939
STAT0004 hybrid	0.069	0.102	0.033	0.424
STAT0004 in-person	0.040	0.094	0.053	0.094

Source: author’s analysis of UCL Moodle log data.

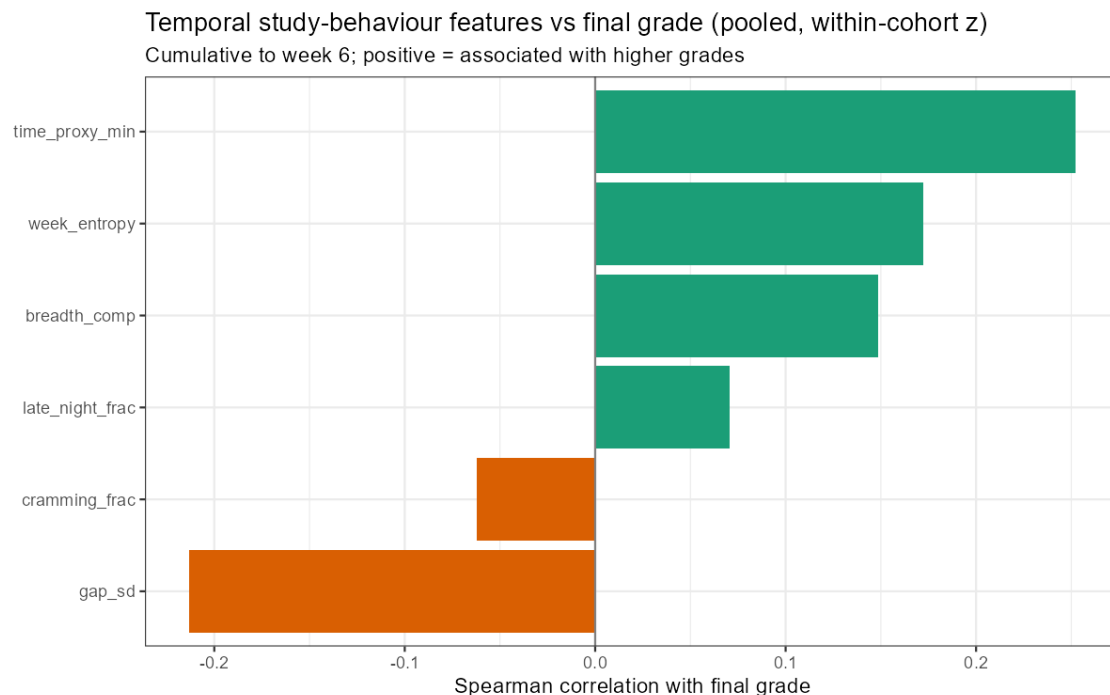


Fig. 8: Temporal and regularity features correlated with final grade.

The diagnostic extensions are not uniform across modules. Temporal regularity adds significant incremental R^2 only in STAT0002, where engagement also aligns more strongly with individually assessed components. In STAT0004, where associations are weaker and group or coursework components carry more weight, the composite and its extensions explain less outcome variance—consistent with greater off-platform study, second-year autonomy, or looser chapter–assessment alignment weakening the Moodle signal rather than invalidating the metric outright.

6.4 Practical limits: lower-tail relevance with modest precision

Quantile regression shows that the engagement–grade slope is steeper in the lower tail of the grade distribution than at the median in every cohort (Table 8, Figure 9). On average, the slope at $\tau = 0.1$ exceeds the median slope by about 2.3 grade points per standard deviation of engagement. Variation in engagement is most consequential among students at greater academic risk—supporting the metric’s relevance for early support—but operational flagging still faces modest precision.

Table 8: Quantile-regression slopes (grade points per +1 SD cumulative IDF).

Cohort	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$	$\tau_{0.1} - \tau_{0.5}$
STAT0002 online	5.32	3.22	2.58	2.11
STAT0002 in-person	7.88	5.04	2.96	2.84
STAT0004 online	6.06	3.15	2.17	2.91
STAT0004 hybrid	3.26	2.66	1.34	0.59
STAT0004 in-person	3.97	1.10	1.28	2.88

Source: author’s analysis of UCL Moodle log data.

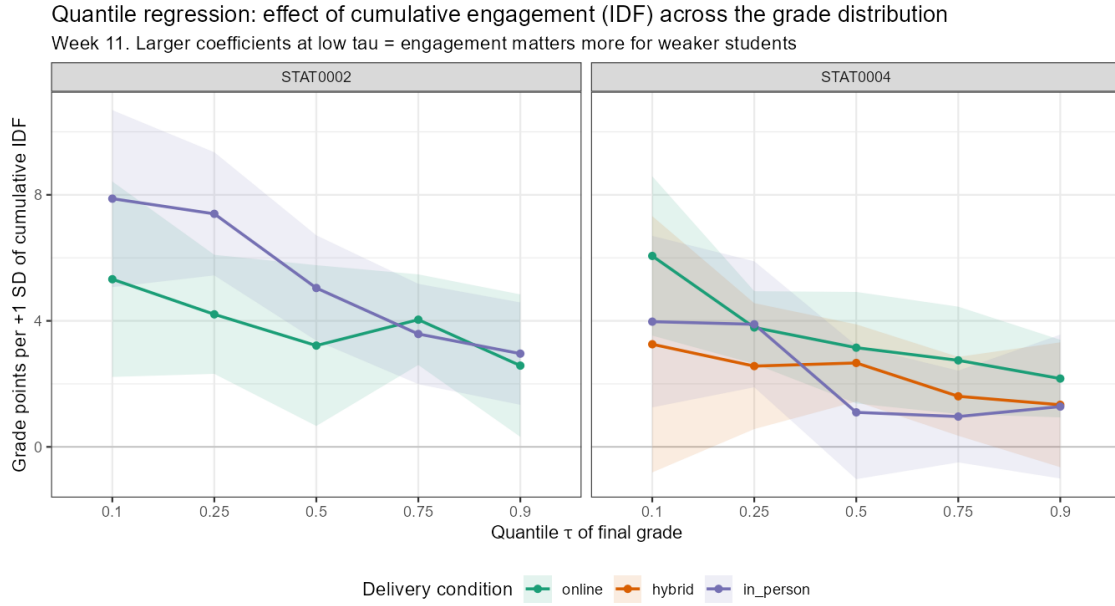


Fig. 9: Quantile-regression slopes of final grade on cumulative engagement across the grade distribution. Slopes are steeper at $\tau = 0.1$ than at the median in every cohort, indicating that engagement variation is most consequential among lower-performing students.

At the course midpoint, the bottom engagement quintile in STAT0002 identifies roughly 38–47% of students who ultimately score below 50% (AUC around 0.71–0.76), but precision is only about 15–26%: roughly three-quarters to four-fifths of flagged students may still pass. Figure 10 illustrates this recall–precision trade-off. The metric is therefore useful for identifying students with low *observed* Moodle engagement, not for identifying students who will fail—appropriate responses are broad, low-cost and supportive rather than intensive or punitive.

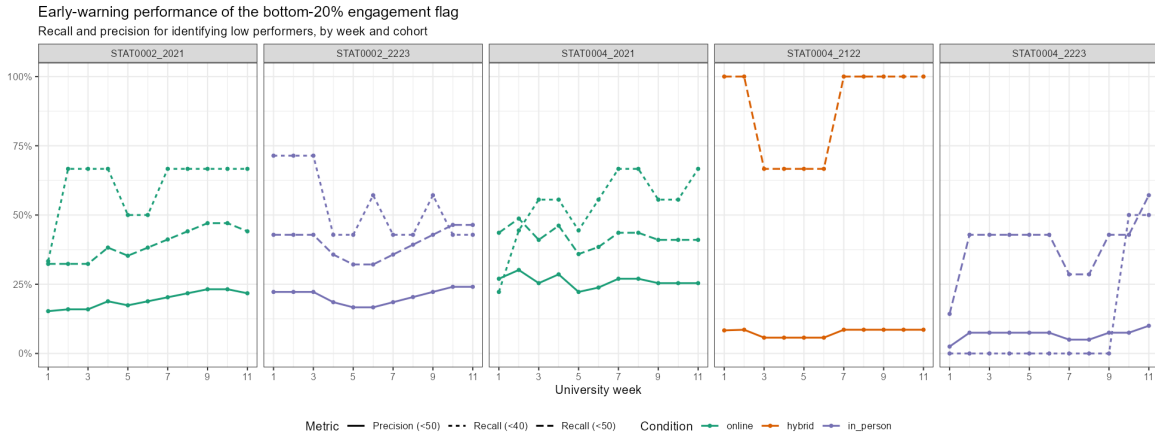


Fig. 10: Recall and precision for the bottom engagement quintile across weeks. Modest precision implies that most flagged students may still achieve passing grades.

6.5 Boundary condition: delivery period and cohort confounding

Because delivery condition is inseparable from cohort and year in this dataset, comparisons across online, hybrid and in-person periods are best treated as a descriptive boundary check rather than a test of delivery effects. Meta-analytic moderator tests do not suggest that delivery label explains between-cohort variation (all $p > 0.87$; Table 2), and weekly correlations appear broadly overlapping across periods (Appendix 11). This should not be interpreted as evidence that de-

livery mode is irrelevant. Module design, assessment structure, cohort composition and Moodle resource organisation may be at least as important as the delivery label itself.

6.6 Robustness checks

Each check addressed a specific threat to the headline conclusions. Programme adjustment tests whether the pooled engagement slope reflects degree-programme composition rather than study behaviour itself; the slope changes by at most 14%, so programme mix is not driving the result. False-discovery-rate correction using the Benjamini–Hochberg procedure guards against spurious weekly associations from repeated testing across weeks and indicators; 205 of 220 tests remain significant. Alternative bounded-outcome specifications address skew and ceiling effects in grade distributions and agree in sign with the main models in all five cohorts. Finally, varying the at-risk engagement threshold examines whether early-warning conclusions depend on an arbitrary quintile cut-off; recall and precision shift in the expected direction without changing the overall interpretation.

7 Discussion

The results support three linked conclusions. First, on **external validation**, the chapter-aligned metric remains a moderate, portable engagement signal in new STAT0002 and STAT0004 cohorts: pooled $r = 0.32$, a mixed-effects slope of roughly 3.8 grade points per standard deviation, positive associations in every cohort, and clear quintile separation in grade outcomes. This extends the primary findings of Johnston et al. (2025) through pooled inference across module-year contexts not used in the original validation study. Stabilisation timing should be read cautiously: early stabilisation in some cohorts reflects a flat, modest correlation rather than strong early predictive power.

Second, on **interpretive depth**, the empirically strongest pattern is that sustained engagement trajectories separate final grades by roughly 9.7 points between low-steady and high-steady profiles—a sixfold difference in the share scoring below 50% (15.7% versus 2.4%). Partial weekly profiles recover the low-steady class with about 81% recall by week 4, making trajectory membership a more actionable mid-term signal than a one-off quintile snapshot, even though precision for eventual failure remains modest. Decomposing the composite adds theoretical depth: once collinearity is accounted for, Diversity and Immediacy carry more unique information than Frequency, consistent with Fredricks et al. (2004)’s distinction between behavioural engagement as structured participation and mere time-on-task. Temporal regularity adds explanatory power in STAT0002 but not STAT0004, and engagement aligns more closely with individually assessed work than with peer- or group-marked components. Together, these patterns indicate that the metric is most informative where Moodle activity maps closely onto individual study and assessment—as in STAT0002—and weaker where students study more autonomously or outcomes depend more on collaborative assessment.

The modules themselves differ markedly in engagement–grade strength, from $r = 0.41$ in STAT0002 in-person to $r = 0.19$ in STAT0004 in-person. Among the plausible explanations—assessment structure, student year and autonomy, and chapter–resource alignment—the assessment-component results provide the most direct supporting evidence: engagement correlates significantly more strongly with the exam than with peer-marking in STAT0002 in-person, and temporal regularity adds incremental explanatory power only in STAT0002 deliveries. That pattern is most consistent with a weaker Moodle–outcome link in STAT0004 because a larger share of the final grade depends on coursework, group work and study away from the VLE, not because the metric fails outright. The parallel with weaker associations in less structured courses reported by Johnston et al. (2025) reinforces this reading: module context should shape local interpretation, not

only whether a pooled association exists.

Third, on **practical limits**, the association is concentrated in the lower grade tail, supporting relevance for early support, but bottom-quintile precision is modest (about 15–26%). The metric identifies students with low *observed* Moodle engagement, not students who will fail. Delivery-period comparisons provide no evidence of a large moderation effect by delivery label, but because condition is perfectly confounded with academic year and cohort, they cannot establish that delivery mode is irrelevant; module design, assessment structure and resource organisation may matter at least as much.

Throughout, the guiding interpretation holds: the metric is a moderate, portable engagement signal; its value lies in diagnostic interpretation rather than raw prediction; and its practical use should be supportive, low-stakes and locally validated.

8 Practical implications

The bottom-engagement flag should be interpreted as identifying students with low *observed* Moodle engagement, not as identifying students who will fail. With precision of about 15–26% at the course midpoint in STAT0002, roughly 74–85% of flagged students may still achieve passing grades. That does not make the metric useless, but it changes how it should be used.

Appropriate uses include broad, low-cost outreach: study reminders, non-accusatory check-ins (“we noticed you may not have accessed much material”), invitations to office hours, prioritising students for human review, and dashboard-level monitoring of chapters and resources. Engagement should serve as a *first-pass* signal, combined with coursework marks, attendance where available, tutor knowledge and student self-report before any substantive intervention.

Inappropriate uses include labelling a student as failing, punitive emails, formal escalation on engagement alone, high-stakes ranking, replacing tutor judgement, or assuming low Moodle engagement means low effort. Low engagement may reflect access barriers, competing commitments or study strategies that do not register fully in logs.

When engagement is low, the more useful question is *why* rather than *whether*: is the student behind the release schedule (Immediacy), touching only a narrow set of resources (Diversity), showing low temporal regularity, or following a persistently low trajectory? From week 4 onwards, partial weekly profiles can assign students to a low-steady trajectory class with roughly 80% recall and 77% precision—a stronger basis for pattern-based outreach (“your engagement has been consistently low across the term so far”) than the bottom quintile alone. Those distinctions guide how support is offered. Any operational use in a new module or delivery context should be preceded by local validation of thresholds and association strength, especially where assessment design or resource labelling differs from the cohorts studied here.

9 Limitations

Several limitations should be kept in view. Delivery condition is confounded with academic year and cohort, so no causal inference about online, hybrid or in-person delivery is possible. The analysis covers only two statistics modules and five deliveries, limiting generalisability. Moodle logs capture online behavioural engagement only; offline study is invisible. Final grades are imperfect outcome measures, especially where group, peer or non-invigilated assessment contributes. Low-performer counts are small in some cohorts, making early-warning metrics unstable. The metric depends on consistent chapter labelling in Moodle. All findings are associational, not causal. Frequency, Immediacy and Diversity are highly collinear, so indicator-level interpretation requires appropriate methods. Finally, external validation would be needed before operational deployment in new contexts without recalibration.

10 Conclusion

This report validates an existing real-time Moodle engagement metric on new STAT0002 and STAT0004 cohorts spanning online, hybrid and in-person delivery periods. Cumulative engagement is moderately associated with final grades, especially among lower-performing students. The sharper findings concern *how* students engage: trajectory classes separate outcomes by roughly 9.7 grade points and can be identified from partial weekly profiles with useful accuracy from week 4; breadth and promptness of resource use outweigh session frequency; and these patterns are strongest where Moodle activity aligns with individually assessed work, as in STAT0002. Delivery-condition comparisons are descriptive only because condition is confounded with academic year and cohort. The metric should be used for low-cost supportive monitoring—broad nudges, check-ins and human review—rather than high-stakes prediction, with thresholds validated locally in each new module context.

A Supplementary figure: weekly correlations by module and period

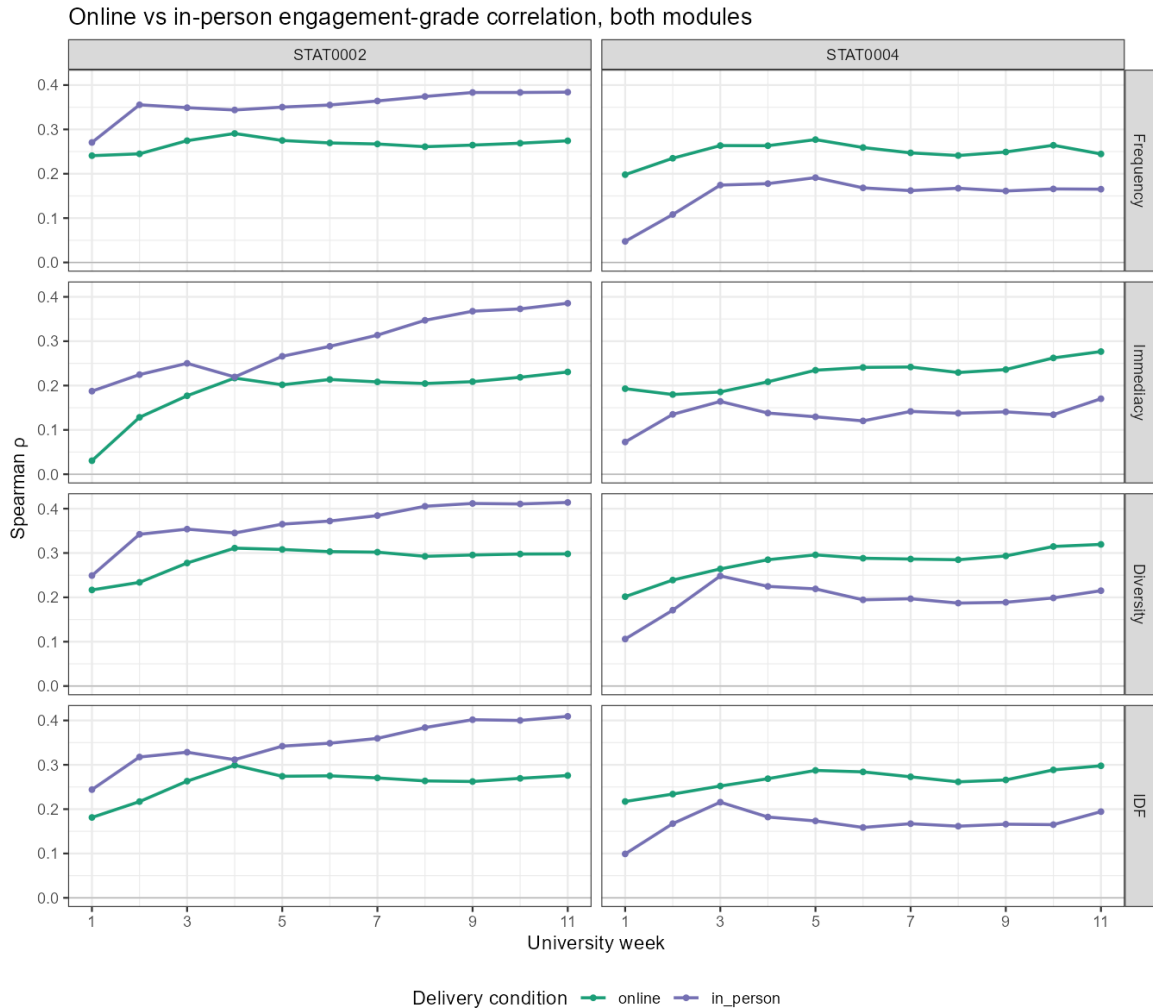


Fig. 11: Weekly engagement–grade correlation by module and delivery period. Comparisons are descriptive; condition is confounded with academic year.

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